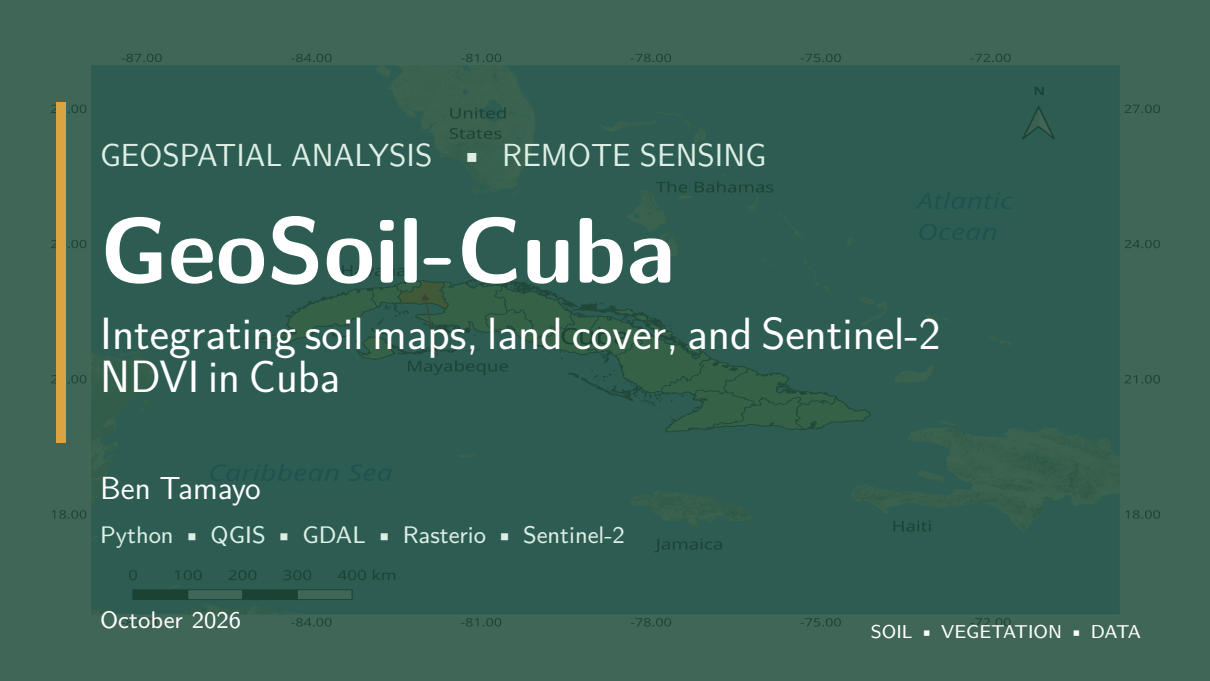


GEOSPATIAL ANALYSIS ▪ REMOTE SENSING

GeoSoil-Cuba

Integrating soil maps, land cover, and Sentinel-2 NDVI in Cuba

Ben Tamayo

Python ▪ QGIS ▪ GDAL ▪ Rasterio ▪ Sentinel-2

October 2026

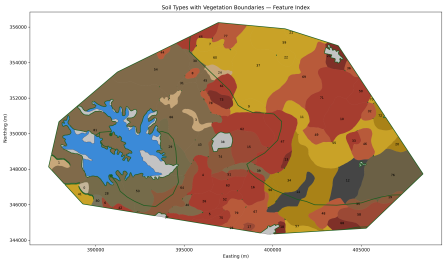
SOIL ▪ VEGETATION ▪ DATA

A GIS workflow built around a real landscape

THE PROJECT

What can we learn by combining historical soil maps with satellite observations?

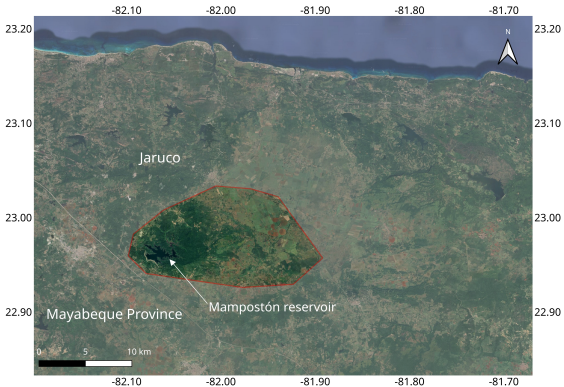
- Prepare historical soil cartography for contemporary GIS.
- Assess spatial coverage and geometric consistency.
- Integrate soil and vegetation information.
- Examine NDVI changes across an agricultural landscape.



Soil units integrated with vegetation boundaries.

KEY INSIGHT The project connects spatial-data quality assessment with satellite-based vegetation analysis.

The study area: Jaruco, Mayabeque, Cuba



Geographic context and location of the study area.

178.47 km^2

Area of interest

91.25%

Reported soil-map coverage

3

Selected potato-field samples

Cuba Norte: EPSG:3795 **Satellite analysis: EPSG:32617** **QGIS + Python**

From historical maps to satellite imagery

DATA AND TECHNOLOGY

Historical soil cartography

Soil polygons, soil classifications, water bodies, and source-map attributes.

Vector data · GeoPackage

Sentinel-2 Level-2A

Red (B04), near-infrared (B08), and Scene Classification Layer (SCL).

10 m spectral bands · NDVI

Land cover and vegetation

Vegetation boundaries and land-cover attributes for spatial integration.

Overlay · Spatial analysis

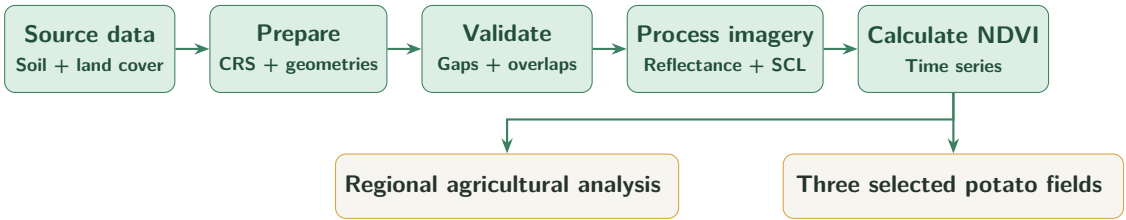
Open-source tools

Python, Jupyter, QGIS, GDAL, Rasterio, and pandas.

Reproducible workflow

One workflow, two analytical scales

METHODOLOGY



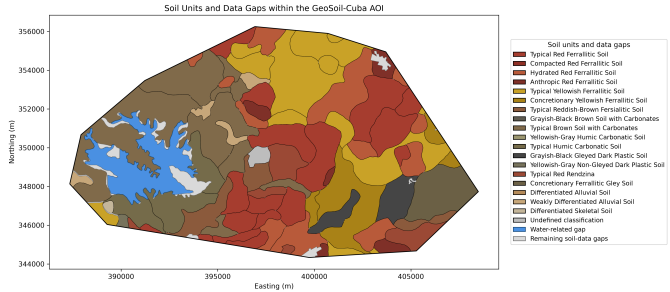
Spatial data quality

Geometry validity, coverage, coordinate reference systems, and overlap assessment.

Vegetation dynamics

Cloud screening, raster calculations, temporal statistics, and field comparisons.

Preparing soil cartography for spatial analysis



Soil units clipped to the area of interest.

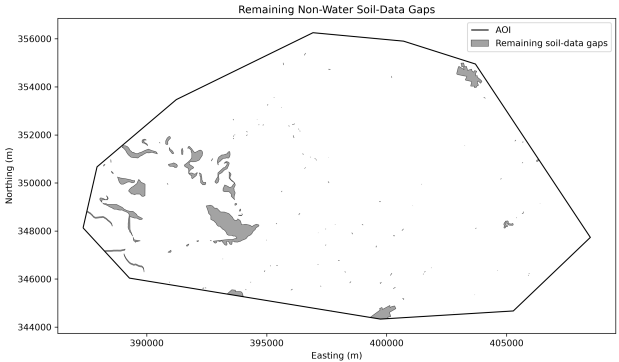
WHY PREPARATION MATTERS

A map is more than its polygons.

- Harmonize coordinate reference systems.
- Inspect geometry validity.
- Evaluate spatial coverage.
- Preserve source attributes.
- Check the consistency of polygon areas.

KEY INSIGHT Spatial analysis depends on understanding the quality and limitations of its input data.

How complete is the soil map?



Spatial distribution of the remaining soil-data gaps.

178.47 km^2

Total AOI

162.85 km^2

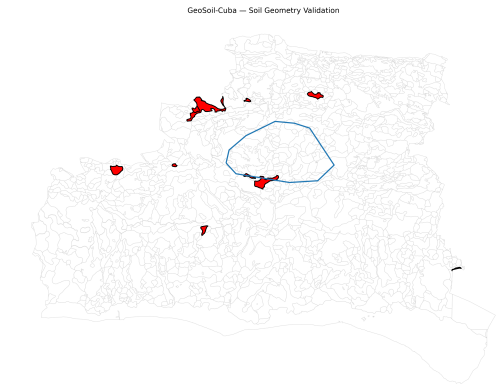
Reported soil-map coverage

4.20 km^2

Remaining non-water gaps

Remaining non-water gaps represent approximately **2.35%** of the AOI.

Geometry checks reveal hidden spatial issues



Geometry inspection and spatial-data validation.

TOPOLOGY RESULTS

90

Polygon pairs with positive-area overlaps

0.054919 km^2

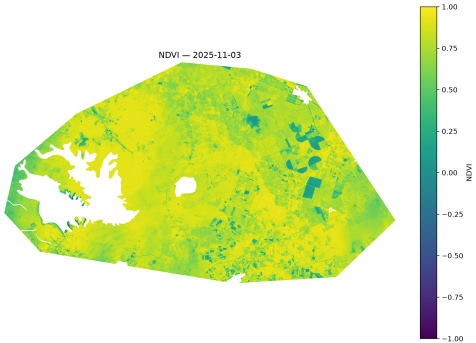
Cumulative pairwise overlap area

0.046807 km^2

Largest reported individual overlap

KEY INSIGHT Area sums and geometric unions help identify overlaps that should be investigated before interpretation.

Turning satellite reflectance into NDVI



Example of the project's regional NDVI processing.

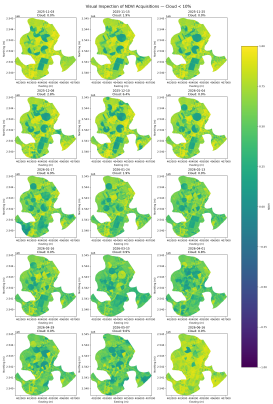
NORMALIZED DIFFERENCE VEGETATION INDEX

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{B08}} - \rho_{\mathrm{B04}}}{\rho_{\mathrm{B08}} + \rho_{\mathrm{B04}}}$$

B08: Near-infrared reflectance
B04: Red reflectance

Higher NDVI values generally indicate stronger vegetation greenness, but NDVI is not a direct measure of crop yield.

Screening satellite observations



Cloud screening and observation selection.

OBSERVATION INVENTORY

226

Initial catalogue observations

32

Processed regional NDVI rasters

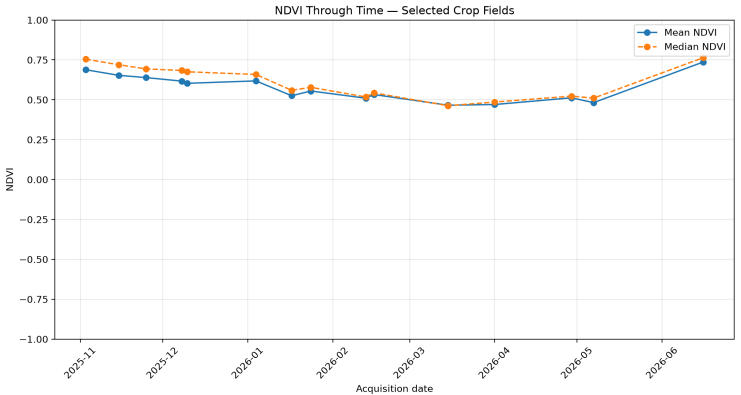
15

Retained regional acquisitions

November 2025 – June 2026

Regional screening threshold: cloud coverage below 10%.

Regional NDVI changes through time



Mean and median NDVI across the selected agricultural region.

MEAN NDVI

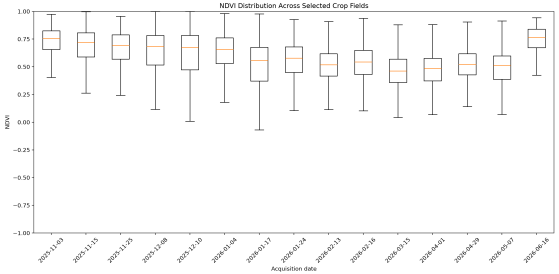
0.688
3 November 2025

0.466
15 March 2026

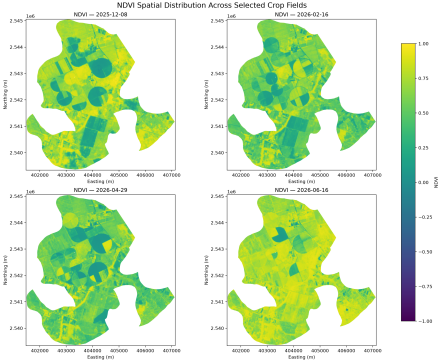
0.737
16 June 2026

KEY INSIGHT Regional NDVI declined from November through March, then fluctuated and increased by June.

One average cannot describe every pixel



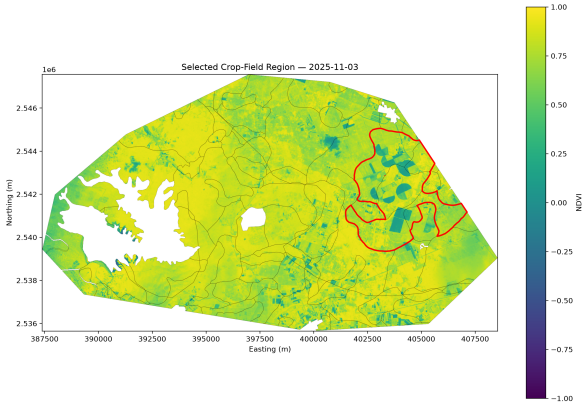
Regional NDVI distributions across acquisition dates.



Spatial variation in NDVI across selected dates.

Distribution plots describe variation within the region; spatial maps show where vegetation-index values differ. Neither alone identifies the cause of those differences.

From the regional landscape to selected fields



Selected agricultural region used for NDVI analysis.

FIELD-LEVEL COMPARISON

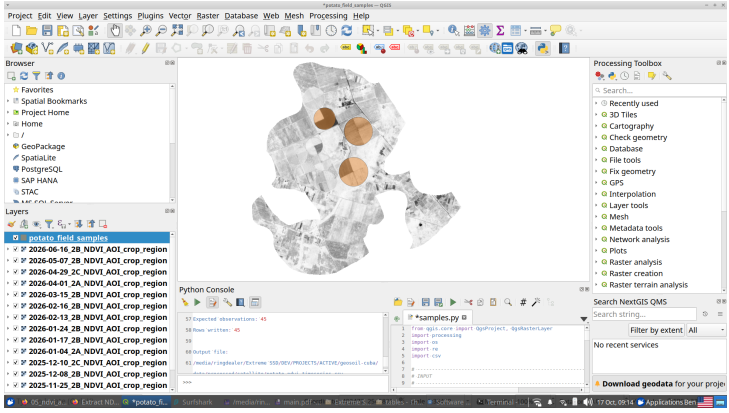
Three locally verified potato fields

- Sampling polygons digitized in QGIS.
- NDVI extracted across 15 accepted dates.
- Field-specific temporal patterns compared.
- Regional averages interpreted alongside local variation.

45

Field-date observations

Three selected potato-field samples



Three sampling polygons displayed over the agricultural-region NDVI raster.

SAMPLING DESIGN

3

Selected fields

15

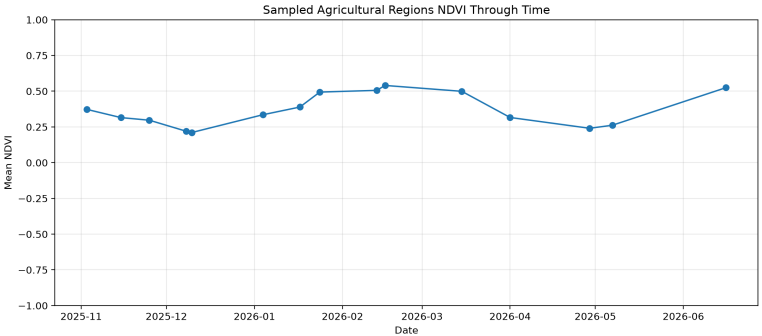
Acquisition dates

45

Field-date observations

Targeted samples, not a statistically representative survey of potato cultivation in Cuba.

Different fields, different NDVI trajectories



Mean NDVI through time for the three selected fields.

OBSERVED MAXIMA

Field 1
0.744
15 March 2026

Field 2
0.690
16 June 2026

Field 3
0.573
24 January 2026

KEY INSIGHT The sampled fields reached their highest observed mean NDVI values on different dates.

What the results can – and cannot – tell us

Supported by this workflow

- Reported soil-map coverage and gaps.
- Detected polygon overlaps.
- Temporal variation in NDVI.
- Differences between selected field samples.

Not established by NDVI alone

- Crop yield or productivity.
- Exact crop phenological stages.
- Causal effects of soil classes on NDVI.
- Representative conditions across Cuba.

Data quality and uncertainty matter as much as visualization.

What GeoSoil-Cuba demonstrates

01

Spatial data quality

Coverage, geometry, and topology should be assessed before interpretation.

02

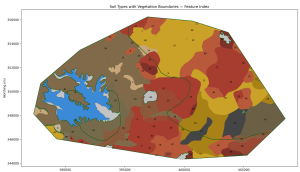
Satellite analysis

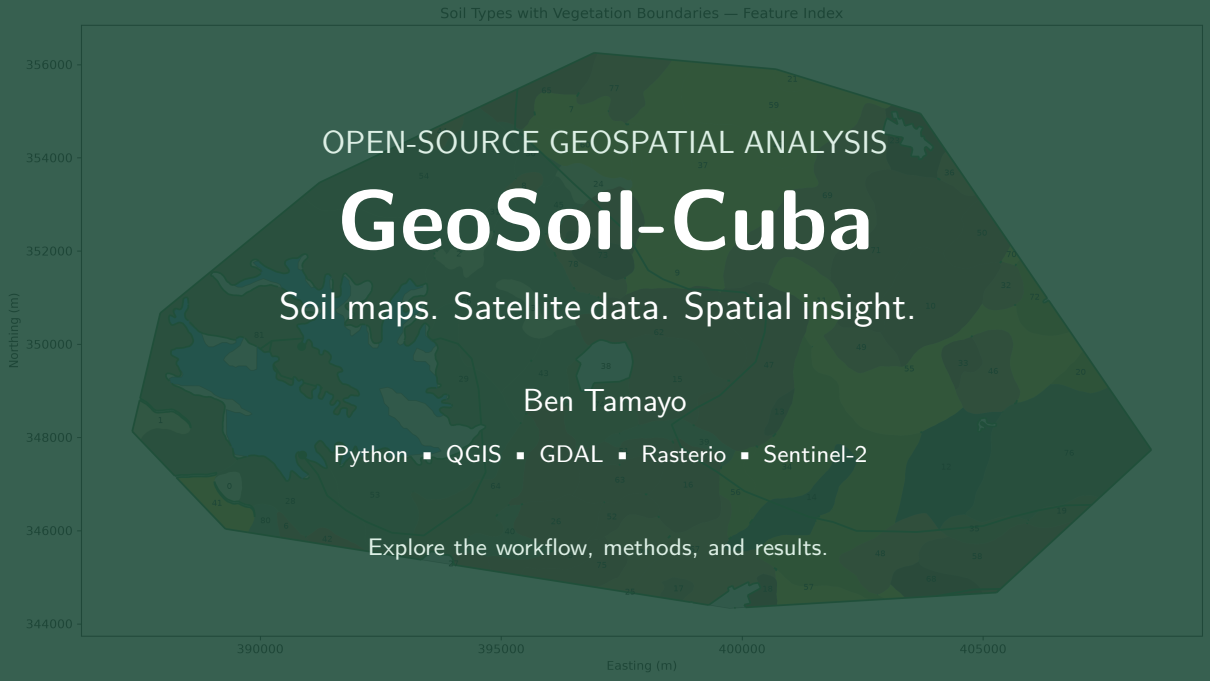
Sentinel-2 NDVI supports the analysis of vegetation-index dynamics.

03

Reproducibility

Open-source tools connect data preparation, analysis, and visualization.





OPEN-SOURCE GEOSPATIAL ANALYSIS

GeoSoil-Cuba

Soil maps. Satellite data. Spatial insight.

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Explore the workflow, methods, and results.